1.4 The Matrix Equation Ax=b

$A = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}_{m \times n} = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}$ $\quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

- Definition:

If A is an $m \times n$ matrix, with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$, and if $\mathbf{x}$ is in $\mathbb{R}^n$, then the **product of A and $\mathbf{x}$**, denoted by $A\mathbf{x}$, is **the linear combination of the columns of A using the corresponding entries in $\mathbf{x}$ as weights**; that is,

$$A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

- Note that ***Ax*** is defined only if the number of columns of ***A*** equals the number of entries in ***x***.

Ex:

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = 4 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + 7 \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$a_1 \quad a_2 \quad a_3$; 2×3 ; 3×1

$$= \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 6 \\ -15 \end{bmatrix} + \begin{bmatrix} -7 \\ 21 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

Ex: For v_1, v_2 and v_3 in $\mathbb{R}^m$, write the linear combination $3v_1 - 5v_2 + 7v_3$ as a matrix times a vector.

$$3v_1 - 5v_2 + 7v_3 = \underbrace{\begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix}}_{x} \rightarrow Ax$$

→ Consider the system $\begin{aligned} x_1 + 2x_2 - x_3 &= 4 \\ -5x_2 + 3x_3 &= 1 \end{aligned}$ As we did in previous section, it is equivalent to

$$x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

vector equation

Using the above definition we can write

$$\underbrace{\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 4 \\ 1 \end{bmatrix}}_{b}$$

This equation has the form $Ax = b$. Such an equation is called a matrix equation.

Theorem:

If A is an $m \times n$ matrix, with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$, and if $\mathbf{b}$ is in $\mathbb{R}^m$, the matrix equation

$$A\mathbf{x} = \mathbf{b} \tag{4}$$

has the same solution set as the vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b} \tag{5}$$

which, in turn, has the same solution set as the system of linear equations whose augmented matrix is

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{bmatrix} \tag{6}$$

A system of linear equations may now be viewed in three different but equivalent ways:

1) as a matrix equation
2) as a vector equation
3) as a system of linear equations

They are solved in the same way by row reducing the augmented matrix.

- The definition of Ax leads directly to the following fact:

The equation $Ax = b$ has a solution if and only if b is a linear combination of the columns of A.

- Existence of solutions:

Is b in $\text{Span}\{a_1, \dots, a_n\}$?

is equivalent to

Is $Ax = b$ consistent?

A harder question:

Determine whether the equation $Ax = b$ is consistent for all possible b.

Ex:

$$A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Is the equation $Ax = b$ consistent for all possible values of b_1, b_2, b_3?

$$\begin{bmatrix} 1 & 3 & 4 & b_1 \\ -4 & 2 & -6 & b_2 \\ -3 & -2 & -7 & b_3 \end{bmatrix} \Rightarrow \begin{matrix} R_2 + 4R_1 \to \\ R_3 + 3R_1 \to \end{matrix} \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 7 & 5 & b_3 + 3b_1 \end{bmatrix}$$

$$R_3 - \tfrac{1}{2}R_2 \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 0 & 0 & b_3 + 3b_1 - \frac{1}{2}(b_2 + 4b_1) \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 0 & 0 & b_3 + b_1 - \frac{1}{2}b_2 \end{bmatrix}$$

Apparently $Ax = b$ is not consistent for every b because some choices of b can make $b_3 + b_1 - \frac{1}{2}b_2$ nonzero.

Def: "The columns of A span $\mathbb{R}^m$" means that every b in $\mathbb{R}^m$ is a linear combination of the columns of A. In general, a set of vectors $\{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ spans (or generates) $\mathbb{R}^n$ if every vector in $\mathbb{R}^n$ is a linear combination of $v_1, \dots, v_p$ — that is, if $\text{Span}\{v_1, \dots, v_p\} = \mathbb{R}^n$.

Theorem: Let A be an $m \times n$ matrix. Then the following statements are logically equivalent. That is, for a particular A, either they are all true statements or they are false.

a. For each b in $\mathbb{R}^m$, the equation $Ax = b$ has a solution.

b. Each b in $\mathbb{R}^m$ is a linear combination of the columns of A.

c. The columns of A span $\mathbb{R}^m$.

d. A has a pivot position in every row.

Theorem: If A is an $m \times n$ matrix, u and v are vector in $\mathbb{R}^n$ and c is a scalar, then

a. $A(u+v) = Au + Av$

b. $A(cu) = c(Au)$

Next week: $Ax = b$ ⇄ $Ax = 0$